American Computer Science League

2023-2024 ● Contest 3: Short Problems ● Classroom Division

1. Boolean Algebra

How many ordered pairs make the following Boolean expression TRUE?

$$A + B(\overline{A} + B)$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. Boolean Algebra

Simplify the following Boolean expression:

$$A + \overline{B + C} + \overline{B} + \overline{\overline{A} + \overline{C}}$$

A. $\overline{A} + \overline{B}$
B. $A + B$
C. $A + \overline{B}$
D. $\overline{A}\,B$
E. $A\,\overline{B}$

3. Data Structures

Find the depth of the binary search tree for:

DODGECARAVAN

A. 2
B. 3
C. 4
D. 5
E. 6

4. Data Structures

If the following operations are done on both a stack and a queue, what is the top of the stack and the front of the queue at the end of the entire process?

```
PUSH("A")
PUSH("L")
PUSH("P")
PUSH("H")
x = POP()
x = POP()
PUSH("A")
PUSH("B")
x = POP()
PUSH("E")
PUSH("T")
x = POP()
```

A. stack top="E", queue front="H"
B. stack top="T", queue front="A"
C. stack top="T", queue front="H"
D. stack top="A", queue front="T"
E. stack top="E", queue front="A"

5. Boolean Algebra

How many ordered triples make this expression TRUE?

$$A(\overline{\overline{B} + C})(\overline{AB + BC})$$

A. 0
B. 1
C. 2
D. 3
E. 4

6. Data Structures

Find the number of nodes that have only 1 child in the binary search tree for:

PONTIACVIBE

A. 4
B. 5
C. 6
D. 7
E. 8

3. Data Structures

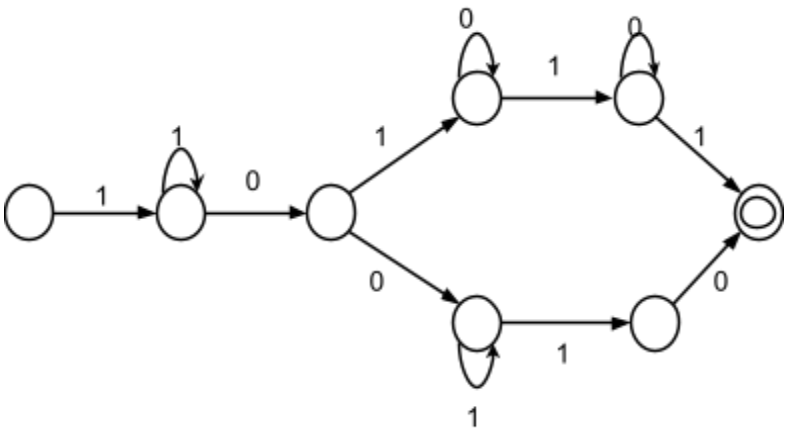
What does the final stack look like from the top to the bottom after performing the following algorithm on an initially empty stack when using the word "ALPHABETS"?

```
for letter = each letter in a word
    if the letter is a vowel
        PUSH(letter)
    else
        x = POP()
        PUSH(letter)
        PUSH(x)
    end if
next letter
```

A. ALPHABETS
B. ESTABAHPL
C. LPHABATSE
D. HPLABASTE
E. STEBAHPLA

8. FSAs

How many of the following strings satisfy the FSA below?



a. 1101011010 b. 11010 c. 1111111

d. 1010101 e. 1100110 f. 1001110

A. 1
B. 2
C. 3
D. 4
E. 5

9. Regular Expressions

Today the pop-up ACSL Coffee Shop is open for business. Only coffee that satisfies the following regular expression may be ordered. How many different kinds of coffee are available?

[b-j] [aeiou] [^k]*

regular	decaf	latte
dark	drip	iced
espresso	black	cappuccino

A. 2
B. 4
C. 5
D. 6
E. 8

10. Regular Expressions

The ACSL Diner opened this year. The cook wants to order seasonings that satisfy the following regular expression. How many will he/she order?

[a-p]* e* [^s]* [d-hp-y]

relish	salt	mustard	pickle
pepper	ketchup	butter	vinegar
mayo	soy	salsa	honey

A. 2
B. 3
C. 4
D. 5
E. 6